

Saltatorial Robotics as a Method of Exploration

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Table of Contents

<i>List of tables and figures</i>	v
<i>Acknowledgements</i>	vii
<i>Abstract</i>	viii
<i>Abbreviations</i>	ix
1. Introduction	1
1.1 Context and motivation	1
1.2 Problem specification	1
1.3 Aim	1
1.4 Objectives	2
1.5 Success metrics	2
1.6 Scope and limitations	2
2. Literature Review	3
2.1 Hopping robots and energy storage	3
2.2 Energy release	4
2.3 Stabilisation	5
2.4 Control approaches	7
2.5 Gap statement	7
3. Design method	8
3.1 System overview	8
3.2 Design evolution	9
Revision A: Motor selection and torque-speed trade-off	9
Revision B: Dual motor configuration and symmetric loading	9
Revision C: release mechanisms	10
Revision D: Ratcheting energy and sawtooth design	11

Revision E: Material selection and tolerancing	11
Design summary	12
3.3 Final mechanical design.....	13
Leg mechanism.....	13
Release mechanism.....	14
Inertial actuators	15
3.4 Electronics and sensing.....	16
3.5 Software architecture.....	17
3.6 Control implementation.....	18
3.7 Test plan and metrics.....	18
4. Results.....	19
4.1 Jump height	19
4.2 Recovery angle.....	20
4.3 Energy recovery	20
4.4 Motor compression.....	21
5. Discussion and Analysis.....	21
5.1 Release efficiency and friction losses	21
5.2 Recovery angle.....	22
5.3 Energy recovery	23
5.4 Objective fulfilment	23
5.5 Environmental considerations for deployment	24
6. Project Management	25
6.1 Timeline.....	25
6.2 Risk register	26
6.3 Budget.....	26
7. Ethical Awareness.....	27
7.1 Safety hazards and mitigations	28

7.2 Responsible testing.....	28
7.3 Sustainability and disposal	29
8. Conclusions and recommendations for further work.....	29
8.1 Conclusions	29
8.2 What this investigation shows	29
8.3 Further work	30
References.....	a
Relevant Appendices.....	f

List of tables and figures

Figure 1- Jumper self-righting sequence by COM displacement. [4].....	6
Figure 2- Annotated image of the final MRV prototype. Final IMU placement differs from image, battery harness now includes a voltage divider circuit for battery monitoring.....	8
Figure 3- Annotated close-up image of the key bearing mechanism. The mechanism is resisting leg rotation hence the bearing is at an angle.	12
Figure 4-(A) Annotated side view of the MRV; (B) Annotated close-up image of the drive mechanism; (C) Annotated CAD section of the drive mechanism; (D) Annotated CAD cross section of the leg and driven gears.	13
Figure 5- (A) Annotated CAD screenshot of the release mechanism; (B) CAD screenshot of inserted connections.....	14
Figure 6- Annotated pewter casting process image.	15
Figure 7- Safety ISR loop, independently from the main loop hosts: red – interrupt timer, yellow – logic, lavender – actuation, light blue – comms.....	17
Figure 8- Software flowchart of the main loop: blue – initialisation, yellow – logic, green – sensor, orange – tool, lavender – actuation, light blue- comms	18
Figure 9- (A) Plotted data of drop height vs spring compression length with a linear trendline. (B) Plotted data of energy recovery efficiency vs possible GPE with a linear trendline.	20
Figure 10- (A) Updated Gantt chart, before exam break. (B) Updated Gantt chart, after exam break..	25
Figure 11- Screenshot of the budget management spreadsheet organised by subassembly.	27
Table 1- Success metrics, includes the measurement, respective objective, target result, mean of actual results.	2
Table 2- Pugh decision matrix for servo energy release including rack, driver gear and driven gear out of the way.....	10
Table 3- Cumulative experimental results table including jump height, recovery angle and energy recovery.....	19
Table 4- Symmetric and unsymmetric wedge compression comparison table across three trials each.	21
Table 5- Project risk register containing name of risk, likelihood, impact, mitigation strategy and post mitigation assessment.....	26
Equation 1- Formula for gravitational torque in terms of mass, gravitational acceleration, CoM height and inclination.....	23

Appendix Fig. 1- Original Gantt chart from PFS, before initial exam break.....f

Appendix Fig. 2- Original Gantt chart from PFS, after initial exam break.f

Appendix Fig. 3- Schematic of the MRV colour coded. Light green: motor wires, dark green: encoder signal, blue: directionality signal, pink: PWM, yellow: signal, red: power, black ground.....g

Appendix Fig. 4-.3D render of the manufactured PCB from within EasyEDA. The PCB design used headers for modularity and DuPont connectivity.....h

Appendix Fig. 5- PCB schematic of the MRV. Blue: bottom layer, high voltage lines, green: ground layer traces, olive: motor output and misc., red: signal wires.....h

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Abstract

The Monopedal Reconnaissance Vehicle or MRV is a prototype jumping robot using dual inertial actuators in orthogonal axes and a spring-based propulsion system. The goal of the project is to develop and demonstrate a novel robotic mobility platform aimed at low gravity exploration. This report outlines the research, design, testing and analysis conducted as a BEng mechanical engineering individual project. Results of this report include research that can form the basis of future deployments such as required energy balances, mechanical and sensory considerations. The final prototype achieved a jump height of 41 cm as a 360 g mass. Using 11.5 cm of compression and a 324 N/m spring, the robot achieved a jump efficiency of 67.8% and 26% landing energy recovery. A maximum single-axis recovery angle of 12.7° was demonstrated using bang-bang control. Deliverables are planned to be published and include all necessary files for reproduction in a github repository under the MIT open-source license. These include CAD, software tools, hyperlinked BOM, detailed design processes and experimental methodology. Results demonstrate that dual-axis inertially stabilised monopedal hopping is feasible, with mechanical refinement identified as the path to improved reliability and energy sustainability.

Abbreviations

MRV – Monopedal Reconnaissance Vehicle

IMU – Inertial Measurement Unit

PID – Proportional Integral Derivative

LQR – Linear Quadratic Regulator

MPC – Model Predictive Control

PFS – Project Feasibility Study

SLS – Selective Laser Sintering

PLA – Polylactic Acid

PCB – Printed Circuit Board

PWM – Pulse Width Modulation

CoM – Centre of Mass

EBS – Engineering Build Space

SFLP – Sensor Fusion Low Power

GPE – Gravitational Potential Energy

EPE – Elastic Potential Energy

PiP – Print in Place

PTFE – Polytetrafluoroethylene

RPM – Rotations Per Minute

MCU – Microcontroller Unit

1. Introduction

1.1 Context and motivation

Jumping is a locomotion strategy attractive for small robotic systems due to its ability to convert stored energy into impulsive vertical thrust, enabling traversal of rough terrain and obstacles without continuous ground contact. In evolutionary terms, saltatorial animals such as the red kangaroo (*Macropus rufus*) achieve up to 70% energetic efficiency by recovering landing energy in tendons for use in the next jump [1]. Inspired by this principle, robotic jumpers can potentially act as efficient and lightweight scouts in environments where wheeled systems are limited by steps larger than the radius of their wheels and flight is constrained by power or atmospheric conditions. This paradigm mirrors the reconnaissance role demonstrated by NASA's Ingenuity, which executed terrain scouting flights ahead of the ground Perseverance rover [2], [3].

However, many currently available mechanisms prioritise take-off performance and mechanical simplicity over attitude control, which leads to poor reliability and low directional control throughout the motion [4], [5]. This project investigates the development of a Monopedal Reconnaissance Vehicle (MRV) that lies at the intersection of bio-inspired efficiency and modern control theory. This single leg jumping robot includes elastic energy storage in the form of a spring and uses dual inertial actuators to stabilise the body about two orthogonal axes. This project evaluates whether controllable, repeatable monopedal hopping stabilised by dual-axis inertial actuators is technically feasible.

1.2 Problem specification

The following specification was derived from the conducted project feasibility study (PFS):

Build a monopedal robot stabilised by two orthogonal inertial actuators, with the ability to perform a repeatable launch, store a portion of landing energy and release it for the subsequent jump. Develop and implement a controller to perform attitude stabilisation and jump actuation of the robot to evaluate the robot using the success metrics in Table 1.

The PFS established a target jump height of 1 m, estimated robot mass of 150 g, a spring constant of 294 N/m and a 10 cm compression length. These parameters were the starting point for the design.

1.3 Aim

The aim of this project is to evaluate the technical feasibility of controllable, repeatable monopedal hopping stabilised by dual-axis inertial actuators. This includes the design and manufacture of an integrated electromechanical platform, development of an energy recovery mechanism and experimental characterisation of jump performance and attitude stabilisation capability.

1.4 Objectives

- O1. Construct an electromechanical platform capable of vertical jumping, validated by Table 1
- O2. Develop a mechanical design that maximises energy efficiency through recovery of landing energy and reduction of friction losses.
- O3. Design and integrate dual inertial actuators about orthogonal axes and demonstrate 2-axis attitude stabilisation.
- O4. Implement and experimentally tune a control algorithm for attitude stabilisation suited to the system's degrees of freedom.

1.5 Success metrics

Table 1- Success metrics, includes the measurement, respective objective, target result, mean of actual results.

Metric	Objective	Target	Mean result
Release efficiency	O1, O2	Maximise	67.8%
Energy recovery ratio	O2	Maximise	26.1-40.1%
Maximum recovery angle	O3, O4	45°	12.7°
Jump reliability	O1	Maximise	57%

1.6 Scope and limitations

The contribution of this report is to outline the development process of monopedal jumping robotics and its interface with dual inertial actuators as the attitude control method. For control, the MRV is treated as two modes: stance (leg is in ground contact) and flight with impact treated as a disturbance. Control is implemented experimentally using a tuned controller rather than derived from a first-principles dynamics model, with model-based approaches such as LQR recommended as future work. Success of this project would yield key functionality like a stable stance at varied leg extensions, repeatable hop execution, landing energy recovery and respective measured data recorded in Table 1. Development of mechanical structures, electrical systems and controlling software enabling main functionality is considered in scope of the project. While designing the robot, specifications are drawn from its future potential use case in scientific exploration in environments such as the moon. As in all developments aimed at space, simplicity, reliability, cost, form factor and mass are considered and optimised throughout. Deliverables include full BOM, CAD models, schematics, experimental methodology, prototype and test design for future research replication.

Environmental effects are not considered in design but are discussed. These effects include radiation, dust accumulation, terrain variance, thermal cycling and effects of long-term use. Environmental hardening is therefore not conducted. End of life disposal considerations including material recyclability and battery handling are discussed in Section 7. Translational locomotion, path planning or other

dynamic sequences are not attempted however are part of recommended future work along with FE and impact analysis which could potentially improve final functionality. Parameters are estimated from the CAD mass properties and leverage manufacturer datasheet readings. Tests include physical constraints such as bearings for 1-axis stabilisation and cable capture for jump tests which may affect the readings and introduce uncertainty that is acknowledged and minimised where possible.

The project was proposed and approved through a PFS submitted in Term 1, which established initial objectives, budget, timeline and component specifications. Given the budget and the overall 24-week schedule with 4 weeks allocated to integration, testing and write-up, priority is given to demonstrable stabilisation capacity and quantifiable jumping.

2. Literature Review

Jumping as a locomotion strategy is not a novel idea with research by NASA into hopping transporters for lunar exploration citing dates as far as 1967 [6]. A similar research question to this report's has been posed as the original Lunar Pogo Stick concept in 1967 where its use was considered for human transport [7]. Benefits of hopping such as energy efficiency, exploitation of weak gravity and independence from smooth terrain are highlighted in research of The Lunar Hopping Transporter [8]. Paradigms such as energy recovery and stabilisation are discussed as upon landing a piston is discussed to partially compress with energy replacement through pneumatics [8]. This parallels the energy recovery and replacement approach investigated throughout this report. These ideas were aimed at being sent with the Apollo missions, similar to this project's targeted Artemis program [9]. Notably, these research papers were heavily limited by their reliance on human command in the absence of current advanced computational, sensor and actuation suites.

2.1 Hopping robots and energy storage

Hopping or jumping is a movement that utilises impulsive, intermittent ground contact to exert control and impulse to enter a desired flight phase. In contrast, walking gaits or roving can exhibit a force throughout the motion by staying in ground contact and having the ability to actuate at any point in time. Multiple methods to achieve hopping or jumping exist and are as follows.

Pneumatics, leverage gas compression and pressure forces in order to provide the necessary impulse to propel the robot. The research paper that inspired projects like the Jax [10] by Marc Raibert for 'Dynamically stable legged locomotion' [11] heavily rely on compression propulsion, usually with a switch on the foot of the robot controlling actuation. This actuation method is attractive due to its high peak power and impulse density, determined by stored pressurized gas. In addition, simplicity of actuation such as a valve enables efficient release of stored potential energy. Implementation of a gas

storage system is its main drawback. Where electricity can be slowly built back up using a solar panel, an autonomous robot that runs out of compressed air is immobilized until a manual recharge. Complexity in pipefittings and safety arise from practical execution of the propulsion method and therefore is not implemented [12].

Inertial impulse hopping, achieved by instantaneously decelerating a spinning disk, is another method to achieve a jump. The angular impulse is translated along the body of the robot to a pivot point that swings the robot into air. This has been demonstrated in field by MASCOT on the Hayabusa 2 mission on an asteroid for data collection and as a proof of concept [13]. Attractiveness of this design stems from the shielding that comes with a fully enclosed robot without exposed mechanics and reduced complexity. However, the robot does not exhibit a controlled nature and samples the surface at random with a stated minimum of two hops. Therefore, inertial hopping is not to be used as the propulsion strategy.

Furthermore, elastics have been used in similar applications due to their inherent high energy density and implementation simplicity. Notable example of this is the record-breaking Jumper by UC Santa Barbara that uses engineered carbon fiber and polymer composite rods under elastic buckling to store and release energy launching the robot 30 m vertically [4]. SALTO is another highly agile monopedal robot that relies on elastic energy storage in the form of a torsional polymer spring and actuation through an 8-bar linkage, translating rotation to a vertical impulse [14]. Given the inherent energetic advantages, elastics are an attractive choice for a small-scale, lightweight reconnaissance robot. In addition, elastics such as springs, exhibit a predictable nature through linear approximations like Hooke's law and therefore can be modelled and hence sourced to a specification. This is the chosen strategy given its feasibility and low risk due to the abundance of suitable prefabricated springs on supplier websites such as Lee Spring [15].

2.2 Energy release

Following storage of potential energy, a mechanism must exist to release it in a sudden controlled burst. Pneumatic solutions use valves connected to external compressed gas supplies. These are limited by the flowrate and require almost instantaneous opening, otherwise the robot does not experience a large enough acceleration to enter flight and instead simply displaces upwards. Approach of the Jumper is to use a winched string system that upon fully loading, physically triggers a release that drops and unwinds the tensioned cable at once. This means that the robot must wind to the limit to execute a jump as the latch does not trigger on command, making it a binary energy release mechanism. To remedy this, it would be possible to add a small actuator such as a servo to open the latch on command. However, this extra complexity and mass would surely reduce the maximum jump height. SALTO does not use a latch and instead relies on timing torque output from a brushless motor during ground contact to further

tension the spring during ground contact. This requires fast control loops and a custom polymer spring attached to an encoder to measure its deflection. While this is a highly mass efficient design and allows a dampened landing through precise motor control, the robot must build up height in a series of jumps in order to reach its maximum height each time, wasting energy. For a reconnaissance mission requiring stationary measurements, this continuous hopping cycle would result in energy loss between jumps. It would increase efficiency and therefore range, if storage and on-command triggering of elastic potential energy is designed for.

For high-risk missions like extraterrestrial exploration, where reliability is paramount, a design principle of minimising complexity is held. Where the Jumper excels is in its low part count and minimal design, so that it lands by direct landing impact [4]. Absence of complex control algorithms, IMUs or other stabilisation actuation greatly reduces the number of failure points and limits power consumption to primary function. With this in mind, the design of the MRV energy system must be mechanical, for passive actuation, and aim to use minimal number of components reducing possibility for failure [5].

Based on the available research, the MRV is to be designed with a vertically displacing leg accommodating spring compression. In mechanics, a ratcheting mechanism is able to limit the direction of input displacement to one axis [16]. A mechanism for ratcheting the leg would enable storage of potential energy recovered from landing and result in a stationary stance at the point of largest spring compression. Along with a method of on-command energy release, this would make the MRV a controllable and inherently efficient vehicle for scientific exploration.

2.3 Stabilisation

As discussed previously, the MRV is to utilise a leg for its propulsion. This means that to direct its jump it is required to execute a jump at an angle. Moreover, to record measurements such as photographs or temperature, it would be optimal that the robot remains stationary [18]. Stabilisation, as discussed here, is correction of the robot's body from a trajectory that might result in immobilisation or damage.

Due to flight being a significant portion of a jumping sequence, it is unsurprising to see numerous developments use propelled thrusters to either stabilise such as SALTO [19] or gain altitude like the Hopcopter [20]. In SALTO's case, geometry of the robot and the presence of a large reaction wheel forced the team to use propellers to overcome physical limitations of orthogonal reaction wheels hitting each other. Thrusters are also a mass-efficient actuator, the motor assembly on SALTO that enables roll and yaw control and weighs less than 7 g [21]. For the purpose of this report, propelled thrusters are not considered as a viable stabilisation method due to atmospheric conditions on the moon or similar celestial bodies with low gravity. There remain three other common stabilisation methods: pneumatic

thrusters, ground stabilising sequences such as used by the Jumper and reaction wheel based inertial stabilisation.

Pneumatic thrusters work on the principle of momentum conservation. By storing and releasing high pressure air, the robot moves in the direction opposite to airflow. Similar to propulsion, pneumatics is not considered further due to the need for manual recharging of air canisters and leakage risks [12].

Ground stabilising sequences are movements designed to leverage a structure in order to regain initial position. For example, the self-righting sequence that Jumper utilised is shown in Figure 1. These offer an electro-mechanical solution to the stabilisation problem with the requirement of ground contact. Need for complex control electronics and algorithms are eliminated by designing a structure that can exhibit a passively stable nature. This paradigm governs the design of a significant part of the Jumper robot and does not add the ability to direct itself. Benefits such as energy efficiency, that arise from stabilising as a side effect of loading the spring are overshadowed by the robot always returning to the same position. Given the nature of the landing and inhomogeneous surfaces present in exploratory environments, the robot may end up jumping in unpredictable directions. Therefore, this method is not used in the MRV.

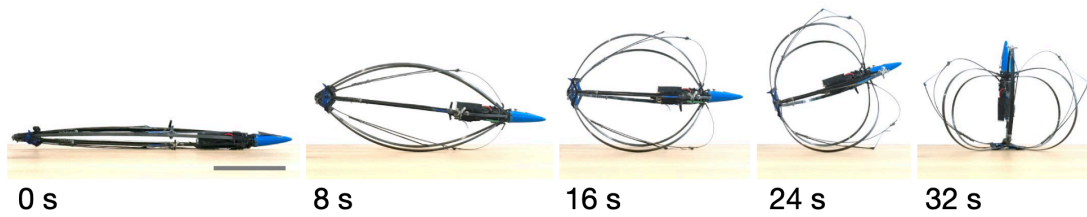


Figure 1- Jumper self-righting sequence by COM displacement. [4]

Reaction wheel stabilisation relies on rotational impulse generated by accelerating masses at a radius. Examples of use include the DEMETER satellite [22] and the CUBLI [23] which use reaction wheels to control their attitude. Given the nature of the chosen propulsion method, an elastically loaded leg, the robot would be considered as an inertially stabilised inverted pendulum, as in the ‘Self-Balancing Stick’ by Mike Rouleau [24] and [25]. This method adds complexity to the mechanism and control algorithms. In addition, sensory electronics such as an IMU and encoders would therefore be required along with more actuators such as motors following current research. A reduction in runtime is expected as stabilisation becomes an active task, in contrast to self-righting, using up more of the battery power. On the other hand, the robot would gain a dynamic controllable nature with a future possibility of remote command execution. Moreover, it eliminates issues regarding ground contact and dependence on atmospheric conditions. Reaction wheel stabilisation is therefore selected as the attitude control method for the MRV.

2.4 Control approaches

Having chosen dual-axis reaction wheels as the stabilisation method, a control strategy for their real-time activation is required. Projects with a similar approach use several control algorithms such as linear-quadratic-regulator (LQR) in CUBLI and Belascuen et al. [23], [26]. This method involves the parametrisation of the system by finding its inertial masses, motor constants and damping factors. This information is then converted into a state-space representation of the system for use in software like MATLAB to derive a LQR seed gain. This process has shown to stabilise a reaction wheel based inverted pendulum in work such as by Witczak et al. [27]. They cover the implementation of LQR and PID into an experimental inverted pendulum set up. The work concludes that LQR offers superior disturbance rejection, though it requires an accurate linearised model of the system.

Model predictive control offers constraint handling and prediction over a horizon but requires solving an optimisation problem at each time step, posing a computational burden unsuitable for resource-constrained embedded systems typical of small-scale and small-budget robotics [28].

Proportional-integral-derivative (PID) is a commonly used controller that involves tuning three values representative of the magnitude, sum and rate of change of the error. Its use eliminates the need to derive complex mathematical models of the robot, instead relying on experiments to hand tune these values. Example of a successful project using PID is Rouleau's Self Balancing Stick, which is a dual axis inertial actuator stabilised inverted pendulum, matching the MRV [24].

The scope of the project is demonstrating feasibility of the mechanical platform. PID allows immediate experimental tuning without complex parameter identification and mathematical model derivation. Unlike the systems studied by Witczak et al., the MRV involves variable inertia with leg extension, nonlinear friction across multiple gear interfaces and backlash in the drivetrain. Accurate characterisation of these parameters within the planned test window is not feasible, making model-based approaches impractical for this project. Moreover, PID has been shown to be successfully deployed on comparable dual reaction wheel systems such as Rouleau's stick discussed earlier [24].

2.5 Gap statement

As discussed in this literature review, hopping robots such as Jumper and SALTO prioritise propulsion without active attitude control. CUBLI and Rouleau demonstrate stabilisation without locomotion. No existing work has been found combining dual-axis inertial stabilisation with elastic energy storage and ratcheted landing energy recovery on a monopodal hopping platform. This project contributes experimental investigation of this combination.

3. Design method

3.1 System overview

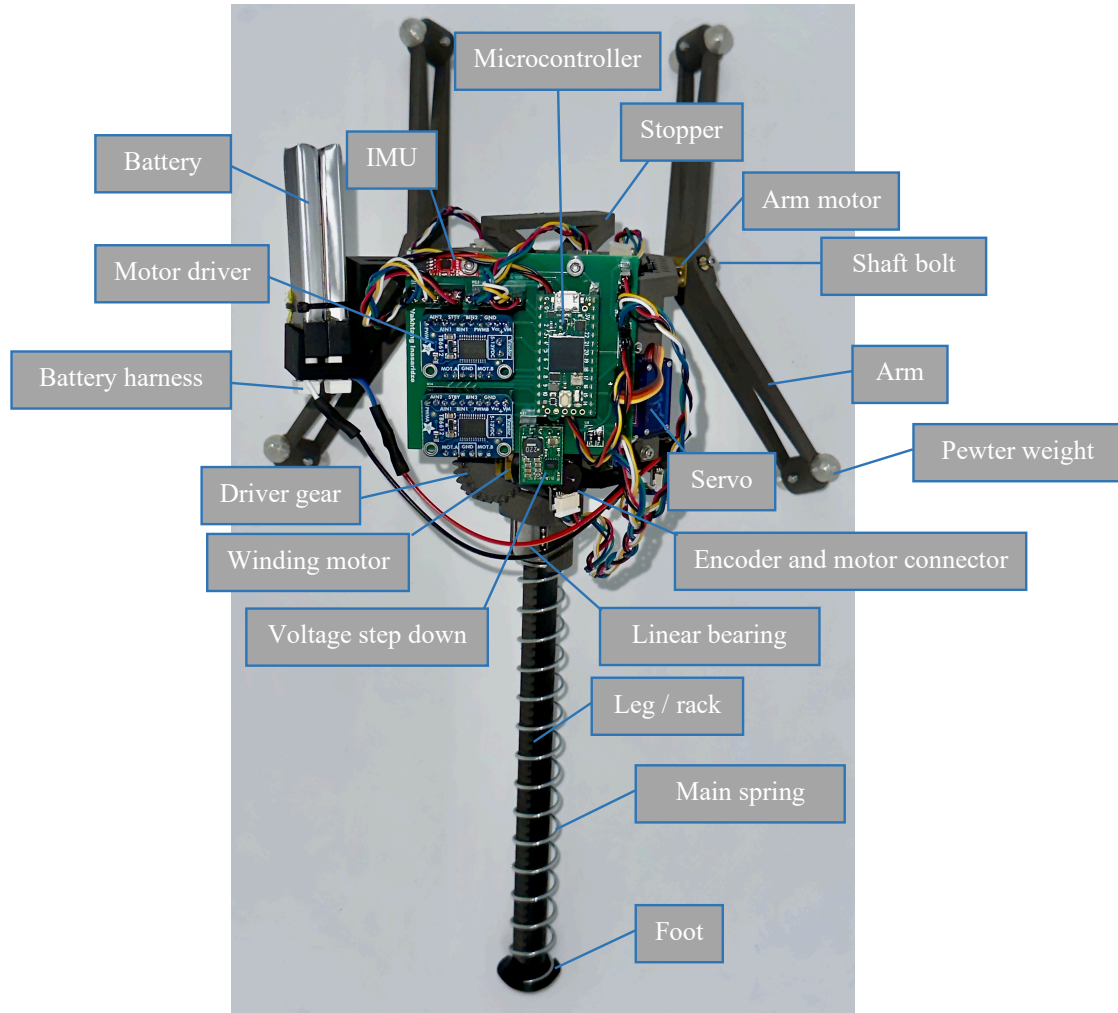


Figure 2- Annotated image of the final MRV prototype. Final IMU placement differs from image, battery harness now includes a voltage divider circuit for battery monitoring.

The MRV is treated as an assembly of three interconnected subsystems: propulsion, stabilisation and control. These are discussed in detail in following subsections and can be seen in Figure 2. The final design of the propulsion system includes the structural components and mechanisms responsible for energy release and recovery such as the leg, gears, ratchet, spring and linear bearing. Stabilisation consists of the two arms with end masses driven by motors to achieve counter rotation in the robot. Lastly, the control subsystem encompasses the electronics that record measurements from the IMU and encoders, execute a control algorithm driving the actuators.

3.2 Design evolution

Revision A: Motor selection and torque-speed trade-off

During the PFS for the proposal of the MRV project, a requirement for a 30N spring force came about. The PFS concluded that 1.5 J of stored elastic energy was required for a 1 m jump at 150 g, achievable with a $k=294$ N/m spring compressed over 10 cm. To accommodate this and convert angular motor torque into linear displacement, a rack and pinion-based design was considered. From the PFS it was also concluded that it was beneficial to maximise compression length x to leverage 2nd order displacement variable in the elastic energy equation $EPE = \frac{1}{2}kx^2$. This would allow the robot to store most energy with the least winding force, reducing internal structural stress and motor torque requirements. A compression length of 10 cm was chosen from the PFS with wind time to be 0.5 seconds, enough to fully compress it during flight. Gears were specced to fit in the robot's approximate dimensions, manufacturing method and circumference requirement. 1.5 cm radius gear was assumed, based on which leg motors could be sourced. Given the assumed gear, the motor must exhibit a minimum torque of 0.45 Nm and 127 RPM to compress 10 cm in 0.5 sec. Given budget and timeline limitations, custom wound motors such as in SALTO were not considered. Brushless and geared DC motors were compared with most cost-torque effective found to be the 1006:1 geared DC motor (COM0802).

The motor fit the torque requirements with a 4 N margin of safety. However, produced 14.2 RPM. At 14.2 RPM with a 1.5 cm radius gear, the motor compresses the spring by 1.1 cm per 0.5 sec flight, storing only 0.018 J per cycle. While the initial jump can take longer to wind up, subsequent jumps maintaining height would require a mechanism with 98.8% efficiency given initial spring energy of 1.5 J. This efficiency is not practically achievable given typical mechanical and motor efficiencies [29].

This demonstrated that gear ratio must be reduced to increase shaft speed at the cost of torque, leading to the adoption of dual, lower ratio motors.

Revision B: Dual motor configuration and symmetric loading

Prioritising shaft speed, the 298:1 (COM0801) motors were selected. The new motors produce 2.5 kg·cm of torque and 72 RPM compared to 5.3 kg·cm and 14.2 RPM of the COM0802 [30], [31]. At 1.2 rps, the motors compress the spring by 5.6 cm in 0.5 seconds, exceeding the 5 cm required assuming 50% energy recovery from landing. Solution to the lack of torque was to use two 801 motors either side of the rack combining to 5.0 kg·cm, comparable to the 1006:1 motor's 5.3 kg·cm. Having driver gears on two sides of the rack also means that the rod's shear forces are symmetric, therefore they negate and reduce lateral frictional losses improving efficiency.

In both revision A and B, the angled gear teeth allow the vertical spring force to push the gears out of mesh. A structure that holds the motors must withstand the sideways force and maintain gear meshing. This results in an increased stiffness requirements and risks material fatigue from force cycling.

With both designs in revision A and revision B, the motors are geared. This means that they produce a significant resistance to free-spin and would limit the release of the leg in terms of speed and efficiency. A solution to this is a release mechanism.

Revision C: release mechanisms

Prioritising simplicity, a missing-tooth gear design was explored first. Essentially, the release would happen when the gears rotated through to a section without teeth to mesh with the rack and therefore allow the spring to extend. Having built and tested a 3D printed prototype, the performance of the mechanism was adequate. The gears meshed with the rod enough to hold a spring with significantly higher stiffness than in the final design. From experiments, it was noted that the spring could only release when a full revolution was completed. This was a binary energy output and meant that the spring released either all the energy or none. This prompted research into variable energy release mechanisms from literature and common mechanisms.

Several alternative release methods were explored including expanding starter mechanisms, one-way bearings, rack displacement and extendable teeth. Each suffered from the same fundamental barrier: the mechanism must simultaneously resist spring force during storage and allow free motion during release. No purely mechanical variable energy release mechanism suitable for the project constraints was identified. Electromechanical solutions were then considered, compiled in the Pugh matrix in Table 2.

Table 2- Pugh decision matrix for servo energy release including rack, driver gear and driven gear out of the way.

Criteria + better, - worse, 0 same	Missing tooth (datum)	Rack out of way	Driver gears out of way	Driven gears out of way
Energy variability	0	+	+	+
Controllability	0	+	+	+
Manufacturability	0	-	-	-
Dynamic coupling	0	-	0	0
Stationary actuator	0	+	-	+
Mech. simplicity	0	-	-	-
Energy efficiency	0	-	+	+
Structural load bearing	0	-	-	+
Total	0	-2	-1	+3

From Table 2 a ‘driven gears out of the way’ release mechanism scored highest and was therefore selected. The driven gear approach combined variable energy release with stationary actuator mounting and structural load isolation, justifying the added mechanical complexity.

Revision D: Ratcheting energy and sawtooth design

Selection of the driven gear release mechanism enabled further energy conservation features. In line with objectives, the mechanism would be benefited by a ratcheting leg. As mentioned previously, this would store recovered potential energy in a passively stable form. Gear directionality could have been implemented by using one-way bearings or a jam rod with a spring. Prioritising mass and budget conservation, one-way bearings were eliminated. This left the spring with jam rod option which would fold the rod in one way, compressing the spring, and push down the jam rod the other way, limiting rotation. This would have required a stiff structure and four more springs, for symmetric rod folding.

To eliminate components, a jamming wedge with compliant print in place (PiP) springs was designed. Cutting down on four springs and reducing the mass of the jamming component. The PiP springs were designed as a single part with the connecting rod between driver and driven gears reducing part count seen in Figure 4 (B). Moreover, is facilitated the wedge to be pushed out of the way in one direction and catch the teeth in the other direction.

Now that the gears ratchet the motors do not do any work holding up the leg while it is in a charged position. However, to store landing energy or ratchet the leg upwards precise servo control from state-estimation to open the gears before landing and catch the robot at its lowest point was considered. This would have required fast feedback and actuation loops, beyond the prototyping capabilities.

A passive solution was once again required to let the leg ratchet upwards. Leveraging knowledge of the lateral forces from leg downward force a teeth design with steep angles was thought of. The idea that given vertical force the teeth would be pushed out of meshing was here utilised. Moreover, the fact that teeth design was now in research, led to the discovery of a shear-less tooth design. In which the lateral forces affecting gear meshing and structural stiffness are eliminated by angling the teeth asymmetrically. This produced a gear teeth design that allows passage when landing, meshes with gears and does not produce lateral forces when winding. Design is seen in Figure 4 (C) and is reminiscent of a sawtooth.

Revision E: Material selection and tolerancing

Throughout prototyping, manufacturing methods and materials have been considered. For initial conceptual prototypes such as the missing tooth, PLA was used. Tolerances and geometric fits were validated using rapid iteration on the Bambulab X1 in the EBS. During preliminary assembly, a snap to

fit leaf broke off when the linear bearing was inserted. Analysing the fault, it was evident that the breakage happened along layer lines. Moreover, when printing the leg, the manufacturing method required it to be split into a two-part print to have a flat build plate contact surface. This manufacturing method led to a concave leg from gluing and forced an increase in part count, such as the stopper being separate. Selective laser sintering (SLS) was therefore adopted, a process that fuses nylon powder layer by layer without supports, producing homogeneous and isotropic parts. Although nylon exhibits lower stiffness than PLA, its toughness and resistance to brittle fracture make it more suitable for the repeated impact loading experienced during jump cycles. Dimensional tolerances differed from PLA, requiring resizing of mounting holes and adoption of M2 threaded inserts for reliable fastening. SLS nylon's isotropic strength enabled the linear bearing leaf, stopper to be integrated directly into their respective parts, reducing component count and eliminating glue joints as failure points. This improved maintainability by allowing the entire leg to be reprinted and swapped without full disassembly.

Design summary

Through five revisions, the design evolved from a single high-ratio motor with a missing-tooth release to a dual-motor, cam-actuated, ratcheting system in SLS nylon. Each revision addressed a specific failure: insufficient shaft speed (A), asymmetric loading (B), binary energy release (C), absence of passive energy recovery (D) and brittle material failure (E). The resulting design satisfies the original objectives of variable energy release, passive ratcheting and repeatable actuation while remaining within the £300 budget. The final mechanical, electrical and software subsystems are detailed in the following sections.

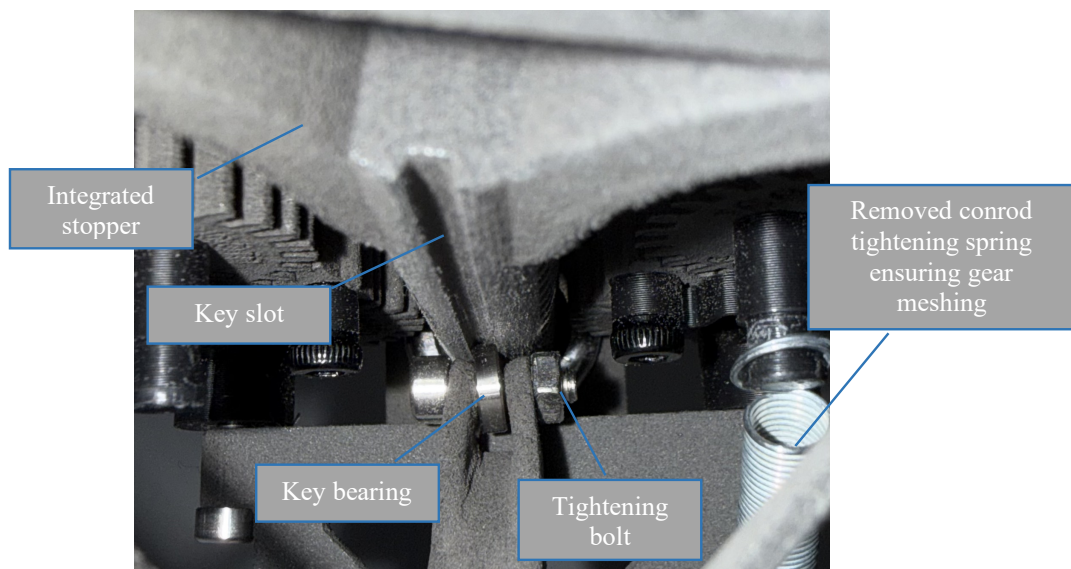


Figure 3- Annotated close-up image of the key bearing mechanism. The mechanism is resisting leg rotation hence the bearing is at an angle.

3.3 Final mechanical design

Leg mechanism

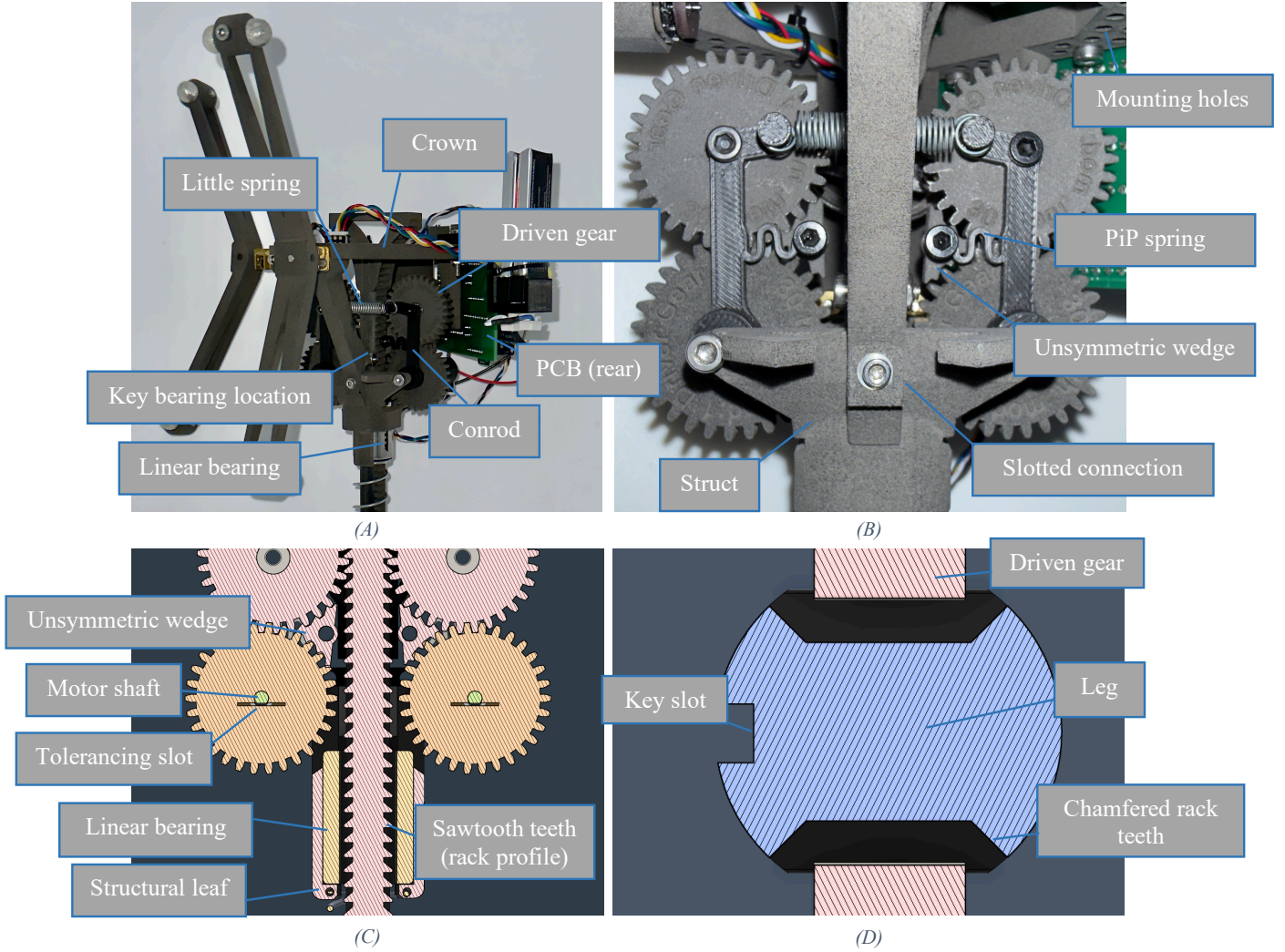


Figure 4-(A) Annotated side view of the MRV; (B) Annotated close-up image of the drive mechanism; (C) Annotated CAD section of the drive mechanism; (D) Annotated CAD cross section of the leg and driven gears.

The propulsion subassembly is predominantly sintered nylon, with PLA retained for the conrods and motor mount. A linear bearing (KH12) was used to ensure reliable and smooth vertical motion of the leg by interfacing at the cylindrical section visible in Figure 4 (A) and (C). Internal diameter of the bearing, 12 mm, constricts the diameter of the leg and the width of the teeth. The maximum compression stroke is 11.5 cm, constrained by the leg length and bearing travel. A key slot, visible in Figure 4 (D), interfaces with a thin key bearing, seen in Figure 3, to prevent rotation of the leg about the vertical axis, maintaining gear alignment. The width of the rack teeth is larger than the width of the gear teeth. This

is to aid re-meshing after a jump given some remaining play in the leg's rotation. The downward angle of the sawtooth teeth allows passive ratcheting on landing while providing a vertical reaction force during winding, as discussed in Revision D. The gear modulus is tuned to be minimal while manufacturable through three iterations. This was to reduce the travel the servo would have to complete prior to releasing the spring. Consequently, this means a shorter time between signal and release aiding efficient actuation timing. The compression spring, visible in Figure 2, has a stiffness of 324 N/m and also serves to clamp the linear bearing leaves around the leg [32]. To prevent the spring from launching the leg away from the main body, the stopper was integrated directly into the leg increasing its strength. The foot is detachable by unscrewing a recessed bolt that connects via a threaded insert in the leg. The spring seats into a recess in the foot, retaining the foot if the bolt loosens.

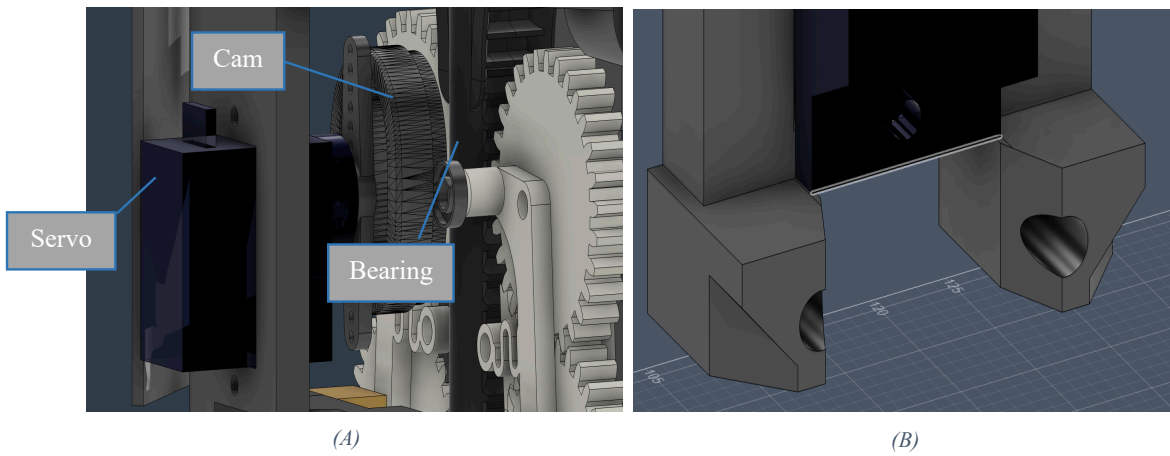


Figure 5- (A) Annotated CAD screenshot of the release mechanism; (B) CAD screenshot of inserted connections.

Release mechanism

As selected in Section 3.2, the release mechanism relies on a 9 g servo. By mounting a cam to the horn of the servo, torque is translated into lateral force on the conrods. When the servo rotates, the cam profile pushes both conrods outward simultaneously, disengaging the driven gears from the rack and releasing the spring. To ensure smooth travel and meshing, thin bearings ride the cam and press against a lip to reduce the possibility of motor shaft bending. Each bearing is bolted down into the PLA conrod and can be seen in Figure 5 (A). After multiple revisions, the servo is mounted vertically with two points of contact. This is to aid in torque distribution from the servo overcoming the friction and slight angle in the driven gear and leg mesh. The shape of the insert, seen in Figure 5 (B), prevents the structure from applying shear force on the bolts that otherwise might break through the connection point wall. This insert approach is replicated at the crown-to-struct interface, creating three points of contact for structural rigidity. The conrods print in under 20 minutes, enabling rapid iteration of the bearing position to tune the mechanical advantage of the release.

Inertial actuators

The inertial actuators consist of two arms at orthogonal axes affecting pitch and roll. At a radius of 79 mm from the motor shaft, the arm is mounted to the shaft with a snug fit and an m2 bolt through a threaded insert digging into a flat portion of the motor shaft. These are manufactured from SLS nylon and have a weight reducing cut out. The arms are angled outward at 25° to maximise the 2nd order displacement from pivot variable in the inertia equation for point masses $I = mr^2$ and prevent arm collision. The arms include safety features such as safety bolts, to prevent masses from sliding out, and chamfered edges. These were included as a contingency to let the arms slide past each other during collision instead of the high impulse of two flat edges, potentially projecting the pewter masses.

The weights were manufactured using a low temperature pewter casting method. As seen in Figure 6, the mould was designed with manufacturability in mind. Made with carbon fiber-reinforced polyphenylene sulfide with a heat deflection temperature of up to 264°C [33]. It was printed with a high density, resulting in a weight of 130 g hence a high thermal mass to absorb heat away from contact surface preventing any mould deformation. The stems connect to the highest point of the masses, ensuring a full filling. A large funnel-like shape was implemented to serve as a buffer for filling, minimise metal spillage and prevent overfilling by being a larger visual sign. Four alignment pins can be seen in the corners of the mould ensuring a symmetric fit of the two-part mould. Rod inserts were designed to be hammered out to prevent complexities of odd object drilling such as custom clamp manufacture. Lead-free pewter was chosen due to its availability as a low temperature casting material, hazard considerations and density similar to that of steel making the volumetric CAD design straightforward. Moreover, it is a soft metal that can be postprocessed using common hand tools in areas such as cutting off the stems, trimming the overflow and filing the edges. Resulting counterweights and arm assembly weighed in at 42 grams, higher than the specced mass of 30 g from the PFS. This overshoot was done in order to ensure enough torque can be applied to the inertia, limiting the required rpm to 7 for a 90° correction. Given the softness of the mass material, unnecessary mass can be tuned rapidly.

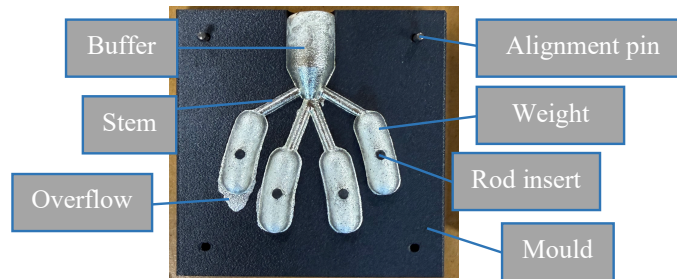


Figure 6- Annotated pewter casting process image.

Powering the arms is a 50:1 geared motor of the same family as the leg motors. Given the motor maximum torque of 3.92 Ncm and the estimated robot body inertia of $5.83 \times 10^{-4} \text{ kg}\cdot\text{m}^2$, the reaction wheel can theoretically correct a 45° disturbance in approximately 0.11 seconds [31]. This provides

substantial control authority for the stabilisation task, with practical performance expected to be lower due to friction, back-EMF and motor speed saturation. As throughout the structure, the motors are snug fit and the rear soldered encoder mechanically prevents any potential for displacement out of the mount.

3.4 Electronics and sensing

Teensy 4.0 – MCU is the core of the control subsystem. Handles I/O of the sensor suite. For example, measures the change in angle per a unit of time to calculate the angular velocity of the arm motors ensuring saturation is not reached. Executes the control loop given input parameters from IMU and outputs actuation to the motor drivers or servo. Chosen for its low profile, low weight and fast processing frequency 600 MHz with option to overclock [34].

IMU (LSM6DSV16X) – a 6 degree of freedom inertial measurement unit, chosen for its high accuracy, onboard low power sensor fusion and small size. At 2.5 grams, the sensor fusion output drifts after high dynamic sequences, e.g. a jump. The yaw axis drifts 5.9° every 5 minutes which was a significant barrier during development and led to a repositioning that aligned the lesser drifting roll and pitch (1.2° , 1.6° per 5 mins) with the arm axes [35]. Drift is associated with the absence of a magnetometer as the chip is specifically specced to 6DoF due to absence of global magnetic fields on celestial bodies like the moon. The chip also features an ML core, uses of which can include jump cycle counting and tracking.

Motor drivers (Adafruit TB6612) – two motor drivers are used, driving two motors each. Per motor, they are capable of delivering 1.2 Amps. Receiving the higher 7.6 V supply directly from the batteries it is possible to modulate down to 6 V by limiting PWM to 77% duty cycle. The breakout board includes three inputs per motor, a PWM and two direction inputs to toggle the H-bridge. PWM is the output of the control algorithm running on the microcontroller [36].

Encoders (PIM604) – shares manufacturer with the motors and provides seamless motor driving and magnetic encoder integration. Solders to the back of the motor using the positive and negative connections. The protruding shaft is used to skewer a disk magnet that is read by the hall effect sensors at 12 counts per revolution [37].

Battery (BetaFPV LAVA) – Two 3.8 V 450 mAh LiHV cells connected in series provide a 7.6 V supply. Given the capacity and the 75 C discharge rate resulting in a maximum current output of 33.8 Amps, well above the required. With its 7.6 V, the battery would store approximately $7.6 \text{ V} \times 0.45 \text{ Ah} \times 3600 \approx 12.3 \text{ kJ}$ of energy enough for hundreds of jumps even at a low efficiency estimate [38].

Stepdown (D24V10F5) – converts the 7.6 V from the battery to a 5 V input for the teensy which then redistributes it to sensors suite and servo. The voltage regulator operates at a minimum 80% efficiency

and supplies up to 1 Amp [39]. This is sufficient as only the logic circuitry and servo are operated through this. Motor power is drawn from respective drivers that have a separate battery connection.

PCB – ordered from a large manufacturer to a specification made in EasyEDA. Hosts all of the components apart from actuators and encoders. Acts as the motherboard for all circuitry. Was completed to save mass from wires, breadboards and solder boards while reducing tangling risk. The PCB is four layers with a ground layer between the high voltage circuitry and sensor or control traces. Given the expected peak currents of up to 900 mA from motors, trace width was adjusted to 30 mils according to page 41 of IPC-2221A with a safety margin and maximum 10 °C temperature increase [40]. Detailed schematic and PCB design can be seen in Appendix Fig. 3, Appendix Fig. 4 and Appendix Fig. 5.

3.5 Software architecture

Control logic of the MRV has been implemented in Visual Studio Code editor. Firmware is built using PlatformIO targeting the Teensy 4.0 with the Arduino framework. The project is structured using a modular file structure where one responsibility is assigned to each file. Architecture of the code consists of five modules: hardware, interfacing the IMU, encoders and motor drivers; math, for quaternion to Euler angle conversion; state, storing and estimating the current robot state; comms, transmitting data over USB serial; main, executing setup and the control loop. All constants are stored in a separate files, config.h and all pin assignments in pinout.h. This was done to ensure centralised correction in case of hardware changes, avoiding logic tampering.

An IntervalTimer triggers a sampling flag at 120 Hz, with I2C reads executed in the main loop to avoid bus conflicts during interrupts. Leveraging IMU built in functionality, the SFLP outputs quaternions at its maximum 120 Hz, hence the flag frequency. The outputs are then converted to Euler roll / pitch / yaw for human readability and control implementation. Encoders read shaft position via hardware quadrature interrupts that are converted to angle and velocity with acceleration for the arms. Safety cutoffs are implemented at 1 kHz independent of the main loop that disable the arm motors via the STBY pin when attitude exceeds 45°. Serial output is sent at 10 Hz for logging and visualisation. Logic flow can be seen in Figure 7 for the safety loop and Figure 8 for the main loop.

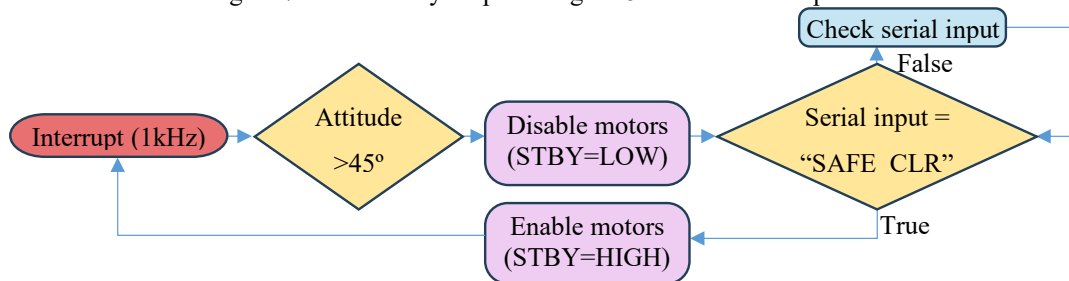


Figure 7- Safety ISR loop, independently from the main loop hosts: red – interrupt timer, yellow – logic, lavender – actuation, light blue – comms

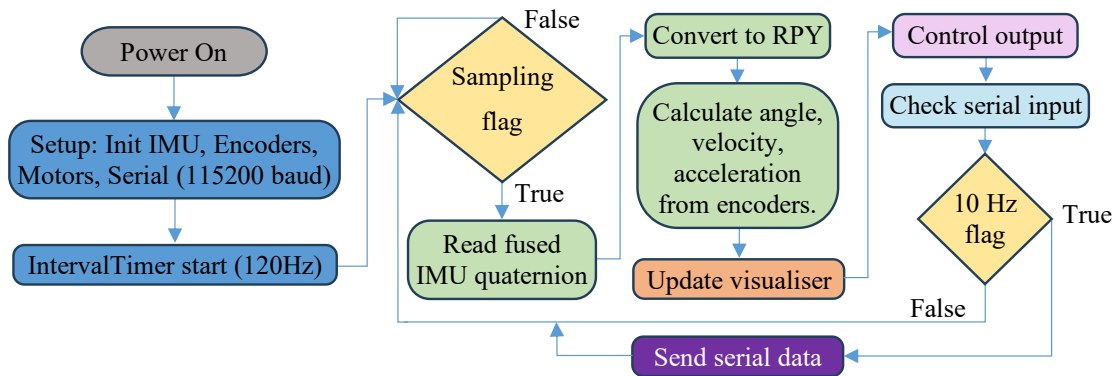


Figure 8- Software flowchart of the main loop: blue – initialisation, yellow – logic, green – sensor, orange – tool, lavender – actuation, light blue- comms

3.6 Control implementation

Following the aims of the project, control strategies demonstrating the efficacy of the mechanical system were implemented. For maximum recovery angle characterisation, a bang-bang controller applying full motor torque in alternating directions was used to establish the upper bound of the hardware's stabilisation capability. A scaling gain proportional to leg extension was applied as the gravitational torque on the inverted pendulum increases linearly with the height of the centre of mass. The controller reads roll and pitch angles from the IMU at 120 Hz and outputs a PWM duty cycle to the respective arm motor driver limited to 77% (6V). PID control was implemented for continuous stabilisation, but tuning was deemed not feasible from lack of control authority. Bang-bang results hence represent the maximum achievable performance of the hardware, which a tuned PID controller would approach but not exceed.

3.7 Test plan and metrics

Test	Setup	Metric	Repeats
1-axis recovery angle	Rig with bearing	Max recovery angle	4 angles
Maximum jump height	Cable capture, metre rule, slow mo,	Release efficiency	4 trials
Energy recovery ratio	Drop from known height, measure compression	Recovery ratio	3 heights

Maximum jump height – determines the largest obstacle the MRV could overcome extending its operating range. A slow-motion camera and meter rule is to be used around the robot performing a jump. A cable attached to the stopper of the robot is to be tightened by an assistant to prevent the effects of ground impact on repeat trials. Maximum displacement of the foot, not including in-flight leg compression, defines the jump height.

1-axis recovery angle – maximum attitude tilts the robot can achieve without falling. The larger the angle, the greater horizontal distance the robot can cover in a single jump with a maximum at 45°. This is to be measured using a rig that constrains the stabilisation to 1-axis and includes angle markings. The robot will start stationary at that angle and aim to accelerate the actuator in order to gain a vertical position at 0°. 2-axis stabilisation was planned but not completed and is recommended as further work.

Energy recovery ratio – amount of energy the motors need to add in order to achieve previous jump height after landing energy recovery as a ratio of initial energy. Simply dropping the robot from varying heights onto the foot and comparing the GPE to the stored elastic energy from a spring compression measurement. The robot is to ratchet the leg upwards and mesh the gears to catch the spring at the point of most compression. Ratio is then calculated as stored energy over available energy.

4. Results

Table 3- Cumulative experimental results table including jump height, recovery angle and energy recovery.

Maximum Jump Height	Maximum Recovery Angle	Energy recovery
42.4 cm	6.8°	from 50 cm: 53.09 mm
40.1 cm	10.7°	from 75 cm: 73.21mm
23 cm	12.7°	from 1 m: 94.42 mm
40.8 cm	14° (fail)	

4.1 Jump height

To measure the efficacy of the spring propulsion system, battery powered jump tests were carried out. The MRV struggled to jump on numerous occasions with the drive mechanism bending from servo actuation without releasing the leg. This is primarily associated with the high static friction in the leg rack and gear meshing. Moreover, the cam mechanism did not reliably manage to release both gears at the same time. From post-test analysis it was the centrality of the cam affecting the displacement of each conrod that led to one of the gears being left on the leg before release. On occasion, as was deduced from the third jump height entry in Table 3, the servo released the leg without providing enough clearance to prevent rubbing. A ratchet like noise was heard during the test which confirmed the cause. Only successful launches were recorded with a reliability of 4 launches out of 7 attempts (57%). Excluding the 23.0 cm trial attributed to gear rubbing, the mean jump height was 41.1 ± 1.2 cm across three successful trials. Given the controlled leg compression of 11.5 cm and the resulting theoretical potential energy limit of 2.14 J, the release reaching 1.45 J of gravitational potential energy with a release efficiency of 67.8%.

During jump tests, the robot exhibited a coupled rotational displacement on launch, visible in slow-motion footage, attributed to asymmetric gear release timing. The tilt was approximately 32° from vertical at the jump vertex.

4.2 Recovery angle

Measurement of the recovery angle was conducted on a rig constrained by two bearings and a foot attachment with a shaft. Bang-bang control was utilised to find the mechanically bound maximum recovery angle. The robot was placed at test angles on the rig; motors were spun up to the upper limit clockwise and then instantly commanded to spin maximally anticlockwise. This generated the largest inertial impulse. The robot started from a stationary angular displacement; this was done to improve experimental controllability ensuring accurate angle measurement. The robot was considered to have recovered if it crossed the vertical equilibrium position. Four angles were tested with a maximal recovery angle of 12.7° and failing at 14° as seen in Table 3. In practice, the dynamic recovery angle during a jump would be significantly smaller as the robot must also overcome existing angular velocity from the disturbance. At 12.7° , the gravitational restoring torque approaches the maximum motor torque of 0.039 Nm, indicating a hardware-limited rather than control-limited result.

4.3 Energy recovery

The robot was dropped from ranging heights onto the leg. Compression was captured by the ratcheting mechanism reliably during all trials and the resulting spring compression was recorded. Three tests of different release heights were conducted to measure the efficacy of the energy recovery mechanism. Plots of which including a linear trendline can be seen in Figure 9 (A) and (B).

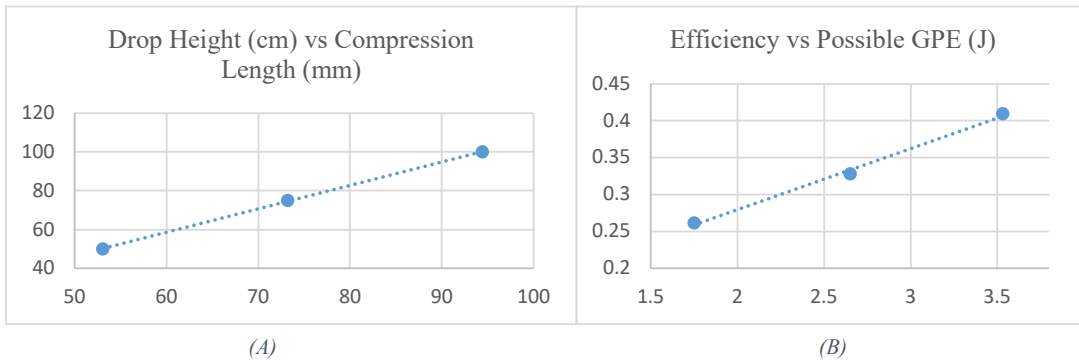


Figure 9- (A) Plotted data of drop height vs spring compression length with a linear trendline. (B) Plotted data of energy recovery efficiency vs possible GPE with a linear trendline.

As seen in Table 3: 50 cm, 75 cm, 100 cm were attempted with respective compression lengths of 53.09 mm, 73.21 mm and 94.42 mm. Calculated efficiencies were 26%, 32.7% and 40.9% respectively. Energy recovery efficiency increased with drop height, rising from 26% at 50 cm to 40.9% at 100 cm. The

robot's own jump at 41.1 cm is closest to the 50 cm test with 26% recovery. This suggests approximately 26% of landing energy would be recovered during normal operation.

4.4 Motor compression

To quantify the effect of incremental efficiency improvements ratcheting wedge geometry was optimised and tested. Symmetric and unsymmetric wedges were tested by commanding the leg motors to wind the spring until an equilibrium position where the ratchet mechanism caught the leg and measurements could be made. Results of three repeats can be seen for each geometry in Table 4, with unsymmetric wedges compressing on average 10 mm more than symmetric. This corresponds to 0.514 J and 0.712 J of stored elastic potential energy for symmetric and unsymmetric wedges.

Table 4- Symmetric and unsymmetric wedge compression comparison table across three trials each.

Symmetric wedge compression	56 mm	58 mm	55 mm
Unsymmetric wedge compression	66 mm	66 mm	67 mm

5. Discussion and Analysis

5.1 Release efficiency and friction losses

The results outlined in 4.1 are attributed to friction in the leg mechanism. Given that it is the only component in contact with the moving leg it must be targeted with lubrication or tighter tolerancing to improve the efficiency number. Another potential source of inefficiency is spring and leg friction. During compression tests and assembly dust accumulation was noticed on the spring. In combination with buckling, that the spring experiences under heavy compression, it is an indicator of frictional inefficiency scraping away at the leg. Both could be improved by using a lubricant suitable for polymers such as silicone grease with PTFE and is recommended for future testing.

Jump reliability at 57% is a result that was expected from the initial design of the double gear release mechanism. A significant cause of this is the cam mechanism that was not able to displace the two gears at the same time. Differences in static friction from initial gear meshing resists the displacement of the cam bending the conrods around the centreline and not releasing. This is worsened when one of the gears successfully de-meshes resulting in the full spring force pressing down on one gear further immobilising it. The gear rubbing is an effect of a slight centrality offset of the cam mounted on the servo; it was fixed by increasing the angular displacement of the servo. The mechanism successfully released in 57% of attempts, demonstrating the concept but highlighting the need for cam refinement.

The prototype is predominantly 3D-printed with experimentally determined tolerances, leaving significant room for dimensional refinement. In particular, improving CoM centrality would reduce the

jump-attitude coupling observed during testing. During jump tests the robot acquired a 32° tilt within 0.28 seconds of launch due to asymmetric gear release. While the rotational kinetic energy is negligible (0.001 J), the resulting attitude exceeds the measured 12.7° recovery angle. This means the robot cannot self-correct after landing without improving the release symmetry and must stabilise in air. Addressing the CoM centrality is therefore critical not only for energy efficiency but for landing survivability.

5.2 Recovery angle

Hardware created the upper bound recovery angle of 12.7° utilising a bang-bang. This is an output of the maximum torque the motors can provide. A tuned PID controller would approach this result but not surpass it. Moreover, the test was completed from stationary meaning no adverse velocity component had to be overcome which is not similar to expected use scenario. Continuing to consider the use case, the recovery angle might be sufficient as the targeted environments experience lower gravity. On the moon where gravitational acceleration is approximately six times lower, the same hardware would recover from substantially larger angles, supporting the intended off-world reconnaissance application.

Although the 32° launch tilt exceeds the ground-based recovery angle of 12.7° , in-flight correction is governed only by body inertia without gravitational torque. The estimated correction time of 0.13 seconds is within the 0.58 second flight window, suggesting that active in-flight attitude control could compensate for launch asymmetry. This estimate assumes instantaneous motor command authority and does not account for reaction wheel momentum reversal time.

The 12.7° recovery angle limits the lateral displacement achievable per jump. At 12.7° from vertical, the horizontal component of the jump is $\sin(12.7^\circ) = 22\%$ of the vertical, meaning for a 41.1 cm jump only 9 cm of lateral travel is achieved. Moreover, any lateral displacement introduces additional energy overhead as the stabilisation system must correct the resulting attitude disturbance on landing, and the recovery mechanism recovers less, reducing the net energy available for subsequent jumps.

Notably, stabilisation while the spring is compressed would benefit from the lower centre of mass, reducing the moment arm and therefore the gravitational torque the actuators must overcome. This suggests that attitude correction should be prioritised during the stance phase before release. During normal operation, the robot is not expected to undergo stabilisation at full extension which gives rise to highest gravitational torque that the actuators have to overcome. The gravitational torque acting on the robot is shown in Equation 1 where L is the centre of mass height above the pivot which varies with leg extension. This linear dependence on L motivated the scaling gain paradigm applied to the control output, increasing actuator authority as the leg extends. At 50% leg extension the centre of mass height halves, reducing gravitational torque and increasing the estimated recovery angle to 25.4° , comparable to the 26° achieved by Belascuen et al. on a dedicated single-axis reaction wheel pendulum [26].

Equation 1- Formula for gravitational torque in terms of mass, gravitational acceleration, CoM height and inclination.

$$\tau = m \times g \times L \times \sin(\theta)$$

5.3 Energy recovery

Figure 9 (B) shows that recovery efficiency reaches 41% at a 100 cm drop height. However, the robot's actual jump height of 41.1 cm corresponds to the 50 cm test condition where efficiency is only 26%. Energy recovery efficiency increases with drop height from 26% at 50 cm to 41% at 100 cm. This suggests a fixed frictional overhead in the ratchet engagement, which constitutes a larger proportion of the total energy at lower impact velocities. Extrapolating, efficiency would continue to increase with drop height but plateau as the fixed overhead becomes negligible relative to the total energy. However, given that the stored energy in the spring cannot surpass the 0.6 m theoretical limit this inefficiency must be eliminated and cannot proportionally minimised.

Key finding from energy recovery results is that the robot would not be able to maintain or increase jump height from initial in its current state. The energy balance reveals that the ratchet recovers approximately 0.377 J per landing while the motors add 0.712 J per winding cycle. Together this accounts for 1.09 J of the required 2.14 J, leaving a deficit of 1.05 J. Reducing the friction losses in the release mechanism would significantly improve this balance. This can be done by optimising frictional assemblies such as the ratcheting wedges. The unsymmetric wedge mechanism stores 0.712 J per motor cycle compared to 0.514 J for the symmetric design, a 38% winding energy improvement from a minor geometric change. This validates the iterative refinement approach and demonstrates that small design modifications can meaningfully impact the energy balance. Improvement opportunities such as these are evident throughout the project such as linear bearing friction or changing the drive mechanisms gear-ratio to produce higher torque. They represent a systematic path to full energetic sustainability as in O2 which the current prototype does not achieve.

Existing robotics discussed in 2.1 do not host passive energy recovery mechanisms relying on constant hopping cycles. The recovery system adds friction losses during jumping. Further work must quantify whether the 26% landing energy recovery outweighs these additional losses.

5.4 Objective fulfilment

O1: The platform was fully designed, manufactured and assembled integrating mechanical, electrical and software subsystems. Testing confirmed vertical jumping at 41.1 cm mean height with 67.8% release efficiency. Launch reliability of 57% indicates the release mechanism needs refinement but the concept is demonstrated.

O2: Energy recovery of 26% was measured with a 1.05 J deficit per cycle. Full energetic sustainability was not achieved but a systematic path through friction reduction was identified. The mechanism demonstrated high reliability with the final iteration having recovered energy of every landing.

O3: Single-axis recovery was demonstrated with a 12.7° maximum recovery angle at full extension. Dual-axis testing was not completed as similar recovery performance is expected, in an offset axis.

O4: Bang-bang control was implemented establishing the hardware upper bound. PID tuning was attempted but not completed and is recommended as immediate further work.

Linking back to the aims and literature, the 41.1 cm jump height at 360 g corresponds to a jump height to mass ratio of 1.14 m/kg. For comparison, SALTO at 98 g achieves 1.25 m at 12.8 m/kg and Jumper at 30 g achieves 30 m at 1000 m/kg [4]. The MRV's lower ratio reflects its dual-purpose design carrying inertial stabilisation hardware and off the shelf components that pure jumpers do not. The 12.7° recovery angle compares to 26° achieved by Belascuen et al. [26] on a dedicated single-axis pendulum without propulsion hardware. The MRV achieves half this angle while also carrying a full propulsion subsystem, spring, ratchet and leg mechanism within the same mass budget. No comparable platform in the literature combines jumping, energy recovery and dual-axis stabilisation, making direct comparison difficult but indicating that the integration penalty is quantifiable and reducible through the improvements identified in this discussion.

5.5 Environmental considerations for deployment

The prototype validates the concept in terrestrial conditions. Deployment in the target lunar environment requires the following considerations:

1. Dust – lunar regolith is abrasive and electrostatically charged [41]. Exposed gears and linear bearing are vulnerable to entrap particles which can act as an abrasive, wearing the mechanism.
2. Thermal – lunar surface temperatures fluctuate between -173°C and 127°C. Nylon becomes brittle at low temperatures [42]. Differential expansion between metal and plastic components will affect tolerancing and might subject the structure to internal stress.
3. Radiation – Teensy 4.0 would require replacement or shielding to prevent damage.
4. Vacuum – no convective cooling for motors. PTFE lubricant outgasses in vacuum [43].
5. Gravity – 1/6th Earth's gravity, as discussed in 5.2, scales up recovery angle and jump height which must be taken into consideration when designing a controller algorithm and IMU tuning.

Despite these challenges, the core design philosophy of mechanical simplicity, passive energy recovery and modular construction translates directly to a flight-hardened version. The bolt-on architecture and standardised threaded inserts would allow material substitution, such as titanium replacing nylon,

6. Project Management



Updated Gantt chart seen in Figure 10 omits a period between 12/12/25 and 19/1/26. This was in order to concentrate on exam preparation for other modules in the Mechanical Engineering degree. The original Gantt chart, from the PFS is included in Appendix Fig. 1 and Appendix Fig. 2. The PFS planned 8 weeks for control theory, 4 weeks for system design, 4 weeks for testing and 4 weeks for report writing. The current date is shown in red bars in Figure 10 (B). Key timeline adaptations include:

- 25

Progress was tracked through weekly supervisor meetings logged on Tabula, with scope and schedule adjustments discussed.

6.2 Risk register

A risk register was maintained and reviewed throughout the project. Key risks and mitigations are summarised in the following Table 5. The register was treated as a living document updated throughout the project as new risks were identified through prototyping. For example, cam asymmetry and IMU drift were not anticipated in the PFS but were added after being encountered during testing. The highest impact risk, not included in the table, was irreversible testing damage. It was mitigated by designing rapidly replaceable components such as conrods that print in under 20 minutes. No irreversible damage occurred during the project, validating modular design as an effective risk mitigation strategy.

Table 5- Project risk register containing name of risk, likelihood, impact, mitigation strategy and post mitigation assessment.

Risk	Likelihood	Impact	Mitigation	Post mitig.
PCB manufacture delay	Certain	Medium	Ordered early, time to order twice.	Medium
Motor torque insufficient for spring	Medium	High	The driver-driven gear ratio is adjustable. Can maximise torque in exchange for RPM.	Low
IMU drift	High	High	Run calibration sequence every trial. Limit trials to 5 minutes.	Medium
Hardware dependency delay	High	Medium	Scope reduced to PID, simulation eliminated.	Medium
Exam period delays	Certain	Medium	Redistributed time after exam, postponed testing window.	Medium

6.3 Budget

A budget of £300 was allocated for the project. Figure 11 summarises expenditure by subsystem. The total spend of £286.66 represents 95.5% utilisation with £13.34 remaining. The largest cost driver was the electronics subsystem at £159.48, driven by PCB manufacture. Materials for 3D printing accounted for £57.83. No budget overrun occurred. The compression spring was obtained as a free sample from Lee Spring, reducing the total cost.

Component sourcing required coordination across five suppliers: Pimoroni for motors and encoders, JLCPCB for PCB fabrication, Pololu for the voltage regulator, Lee Spring for the compression spring and BetaFPV for batteries. Lead times ranged from 2 days (Pimoroni, UK) to 17 days (JLCPCB, China

including the reorder). Managing these in parallel was essential to maintain the critical path, particularly when the PCB fault required a second order while other subsystems continued development.

Budget constraints influenced component selection throughout. For example, off-the-shelf geared DC motors were chosen over custom brushless alternatives, and PCB headers were soldered by hand rather than purchasing the surface mounting service. The £13.34 remaining was retained as contingency for replacement components during testing, such as conrods or wedges that cost negligible material.

MRV LEG									
Part	Link	Price	w. Tax	Delivery	Note	Quantity	SKU	Total	
Spring	link	£ 20.00	£ 24.00	£ -	They don't say the delivery fee until I order?	0	LCM125HK05 M	£ -	
DC geared motor	link		£ 5.20	£ 3.30	GR 298:1	2	75-0639	£ 13.70	
Motor Driver	link		£ 6.70		dual motor controller with separate v input	2	ADA2448	£ 13.40	
Linear bearing	link	£ 7.74	£ 9.29		the tolerance required for the igus is H9	1	KH1228-EU	£ 9.29	
Thin bearing	link	£ 2.23	£ 2.68	£ 2.50	tiny	6	683-EU	£ 18.56	
Servo	link		£ 4.00		connect the 5v and gnd to step down, control at 3.3 from teensy	1	103114	£ 4.00	
Total								£ 58.94	
MRV BRAIN									
Part	Link	Price	w. Tax	Delivery	Note	Quantity	SKU	Total	
Teensy 4.0	link		£ 25.80	£ 3.50	the processing unit	1	DEV-16997	£ 29.30	
Battery	link		£ 22.00	£ 5.90	to power the robot	1	BetaFPV LAVA45	£ 27.90	
IMU	link		£ 24.50		to find cycle position and angle	1	SEN-21336	£ 24.50	
Encoders	link		£ 7.50		magnetic encoders to find tail angular velocity	2	PIM604	£ 15.00	
Qwicc cable	link		£ 1.00		to connect the imu to the teensy solderless	1	104964	£ 1.00	
StepDown	link		£ 12.50		to step down the 2S voltage for logic power	1	POL-2831	£ 12.50	
PCB1		£ 3.73	£ 7.64	£ 15.75		1		£ 23.39	
PCB2		£ 5.30	£ 9.61	£ 16.28		1		£ 25.89	
Total								£ 159.48	
MRV TAIL									
Part	Link	Price	w. Tax	Delivery	Note	Quantity	SKU	Total	
Motor	link		£ 5.20		GR 50:1	2	75-0638	£ 10.40	
Total								£ 10.40	
MATERIALS									
Name		Price per	w. Tax	Column1	Note	Quantity		Total	
PLA		£ 27.06			quantity in kg	0.292		£ 7.90	
Nylon Powder		£ 101.00			quantity in kg	0.3		£ 30.30	
PPS-CF		£ 151.00			quantity in kg	0.13		£ 19.63	
Total								£ 57.83	
								Grand Total	£286.66

Figure 11- Screenshot of the budget management spreadsheet organised by subassembly.

7. Ethical Awareness

Ethics and ethical principles for engineering professionals is a set of required behaviours and practices that guides engineering decision making and taking responsibility for impacts on people, society and environment. These were influential throughout the MRV project by aligning processes, discussions and analyses with ethical guidelines of prioritising safety of the public, such as in the Engineering Build Space (EBS), and rigour in design decision and reporting. Guidelines associated with the location of processes are strictly followed with supervision from competent project supervisor through the weekly project meetings and daily interaction. All source code and CAD files are to be open sourced under the MIT license to enable replication and further development.

7.1 Safety hazards and mitigations

Primary safety concerns of the project are related to unexpected and uncontrolled energy release from either the spring or the battery. In the final design of the robot, with the truncated spring, a full compression of 11 cm stores approximately 2.14 J of elastic potential energy, held in place by 3D printed nylon gears. This imposes a projectile risk if the leg extends without an immediate stopping mechanism. Launching the leg assembly in a potential direction of a user of the EBS breaches 05, 06 and 07 of the Core Rules of the EBS [44] and chapter 2 of the Ethical Principles by the Royal Academy of Engineering [45] hence was mitigated.

As a direct mitigation feature of such risks the final MRV prototype includes the stopper as part of the leg, reducing failure points associated with mitigation strategies. The foot of the leg is detachable and experiences up to 35 N of tension while the gears are closed and the robot is storing potential energy. The risk of a foot projectile is mitigated by screwing down the foot into a threaded insert heat pressed into the core of the leg, well within the FLM printing lowest pull-out force of 1500 N [46].

The reaction wheel counterweights pose a projectile risk if detached during attitude correction. Lead-free pewter was selected partly for its high density, allowing the required moment of inertia to be achieved at lower angular velocities, reducing kinetic energy. Each mass is secured with a safety bolt through the arm mechanically preventing displacement.

LiHV cells are used as the power source of the MRV. Puncturing or short circuiting them poses a fire risk during testing. To mitigate this, a proper battery harness has been manufactured with no exposed conductors. Moreover, the battery is securely attached to the side of the robot and wires do not pass near moving parts preventing entanglement.

Pinch points near the gear mechanisms cause a risk to finger or loose hair in attempts to manipulate the robot during winding. Main drive and motor gear boxes must not be powered when handling the robot and loose hair securely tied back. PPE such as gloves is strongly recommended.

A software safety system detects attitude exceeding 45° and disables motors independently of the main control loop via a 1 kHz interrupt service routine. This prevents uncontrolled arm acceleration while the robot is lying flat preventing breakages resulting in projectile or electrical hazard. PPE such as goggles must be worn.

7.2 Responsible testing

Testing was conducted in the EBS under supervision with appropriate PPE including safety glasses. The robot was operated wirelessly on battery power to eliminate cable entanglement near the mechanism. Jump tests used a cable capture system held by an assistant to prevent uncontrolled ground impact. The

software safety system can be manually re-enabled only via a serial command, ensuring the operator consciously resumes operation. Stabilisation tests are completed using robust axial bearings and desk clamps to ensure the rig doesn't tip over with the robot. Trials are to be limited to 5 minutes maximum to avoid the onset of IMU drift jeopardising stabilisation effectiveness. For drop tests, batteries were removed and replaced with equivalent mass to prevent impact damage to live cells.

7.3 Sustainability and disposal

Sustainability and disposal have been considered in the design, manufacture and sourcing of components. For example, all PLA parts were optimised to use as little to no supports, minimising waste. SLS nylon powder used as support material is partially reusable or recycled by the manufacturer. Lead-free pewter was used in compliance with safety and environmental standards. LiHV batteries are to be disposed or recycled through university battery recycling [47]. PCB was manufactured by JLCPCB in compliance with RoHS 2.0 standards [48]. Structural components such as the crown use standardised mounting holes, allowing individual parts to be replaced without reprinting the full assembly.

8. Conclusions and recommendations for further work

8.1 Conclusions

1. A monopedal jumping robot with dual-axis inertial actuators and ratcheted energy recovery was designed, manufactured and tested.
2. The robot achieved a mean jump height of 41.1 cm with a theoretical maximum of 60.6 cm, corresponding to a release efficiency of 67.8%.
3. The high reliability ratcheting mechanism recovered 26% of landing energy at jump-representative heights, with efficiency increasing to 41% at higher impact velocities.
4. Maximum recovery angle of 12.7° was demonstrated on a single axis using bang-bang control, representing the hardware upper bound. At 50% leg extension it increases to an estimated 25.4° .
5. Jump reliability of 57% was traced to cam asymmetry in the release mechanism and leg bending, identified as the priority improvement.
6. The 32° launch tilt from asymmetric release exceeds the ground recovery angle but is theoretically correctable within the 0.58 second flight window, motivating development of state estimation for use in the gain scheduled controller.

8.2 What this investigation shows

This investigation demonstrates that controllable, repeatable monopedal hopping stabilised by dual-axis inertial actuators is achievable, with identified paths to improved reliability and energy sustainability. The project validates the integration of elastic energy storage, ratcheted recovery and reaction wheel

stabilisation on a single platform, a combination not found in existing literature. The identified failure modes are mechanical in nature and addressable through iterative refinement rather than fundamental redesign, indicating that the concept is sound. The primary outcome of this project is the prototype itself and the demonstration that dual-axis inertially stabilised monopedal hopping is feasible. Performance metrics serve to characterise the first prototype and identify improvement paths, not to optimise a final design.

8.3 Further work

All files are to be open sourced under the MIT license for replication. The author recommends the following list of significant suggested further work ranked for anyone continuing the MRV project:

1. Leg linear encoder implementation. A key finding of this project is the necessity of precise leg extension knowledge for gain-scheduled control. This makes the key bearing, preventing the leg rotation about vertical axis, essential for this paradigm to prevent encoder misalignment.
2. LQR controller experimentation. With the implementation of the leg encoder, precise knowledge of the CoM is at hand and can be implemented in the model-based controller. To complete this point, one must also conduct work with parameter identification making it a significant undertaking and representing a scope beyond the 24-week project timeline. Within this, complete PID tuning for continuous stabilisation, using the bang-bang recovery angles as performance targets.
3. Translational locomotion and path planning. This is perhaps the most challenging undertaking suitable for PhD research of a time and financial scale similar to SALTO, requiring years of mechanical perfection and control development.
4. Parallel axis dual inertial actuators for stabilisation. Instead of orthogonal axis, another possible stabilisation method would be to place the axis parallel. This would remove the geometric constraint on reaction wheel radius imposed by orthogonal arm clearance, potentially reducing total robot mass. Yaw control of the robot would be an additional feature that could be used for camera panning. The author suggests this project for anyone interested in characterising a completely new method of inverted pendulum stabilisation perhaps as a research paper similar to the ones in the literature review.
5. FE and impact analysis. Structural simulation of landing loads on the designed structure. This would produce significant material savings throughout the structure through topology optimisation.
6. Environmental hardening. As the aim of the MRV is to explore celestial bodies such as the moon, protection against radiation, thermal cycling and dust sealing must be completed. This is ranked lowest due to being the final step of MRV development.

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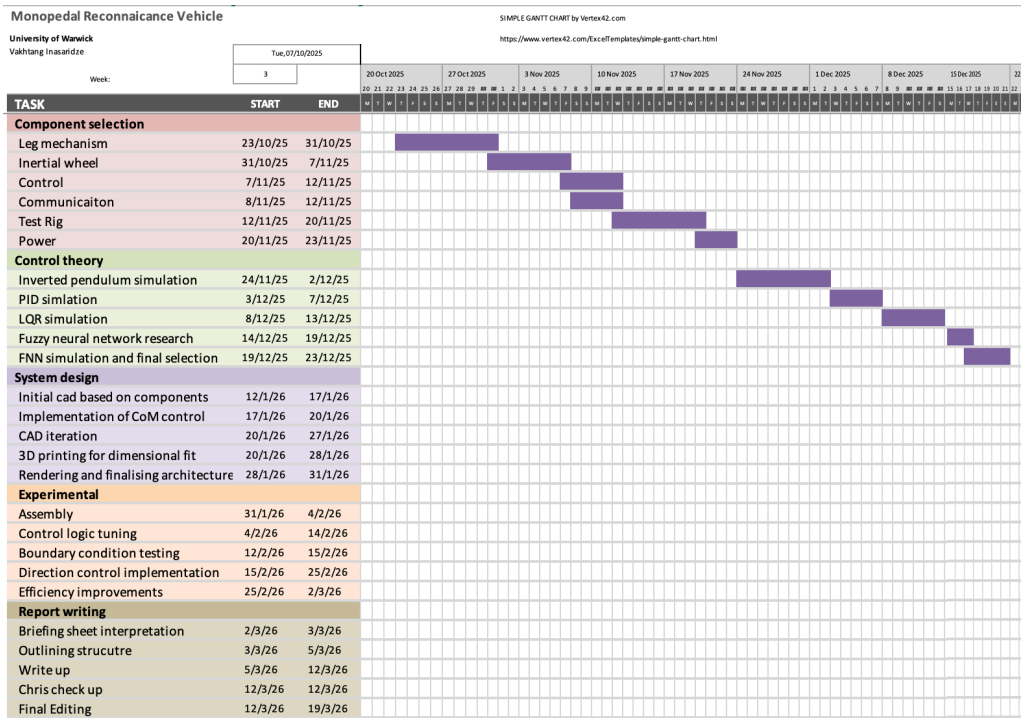
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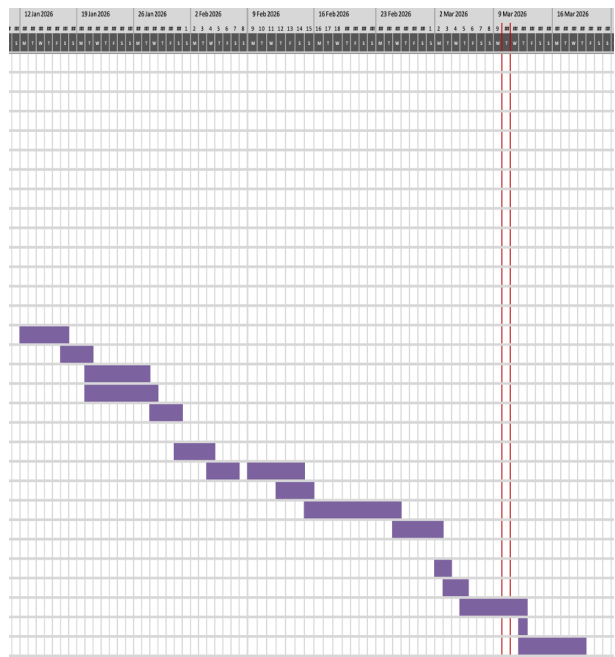
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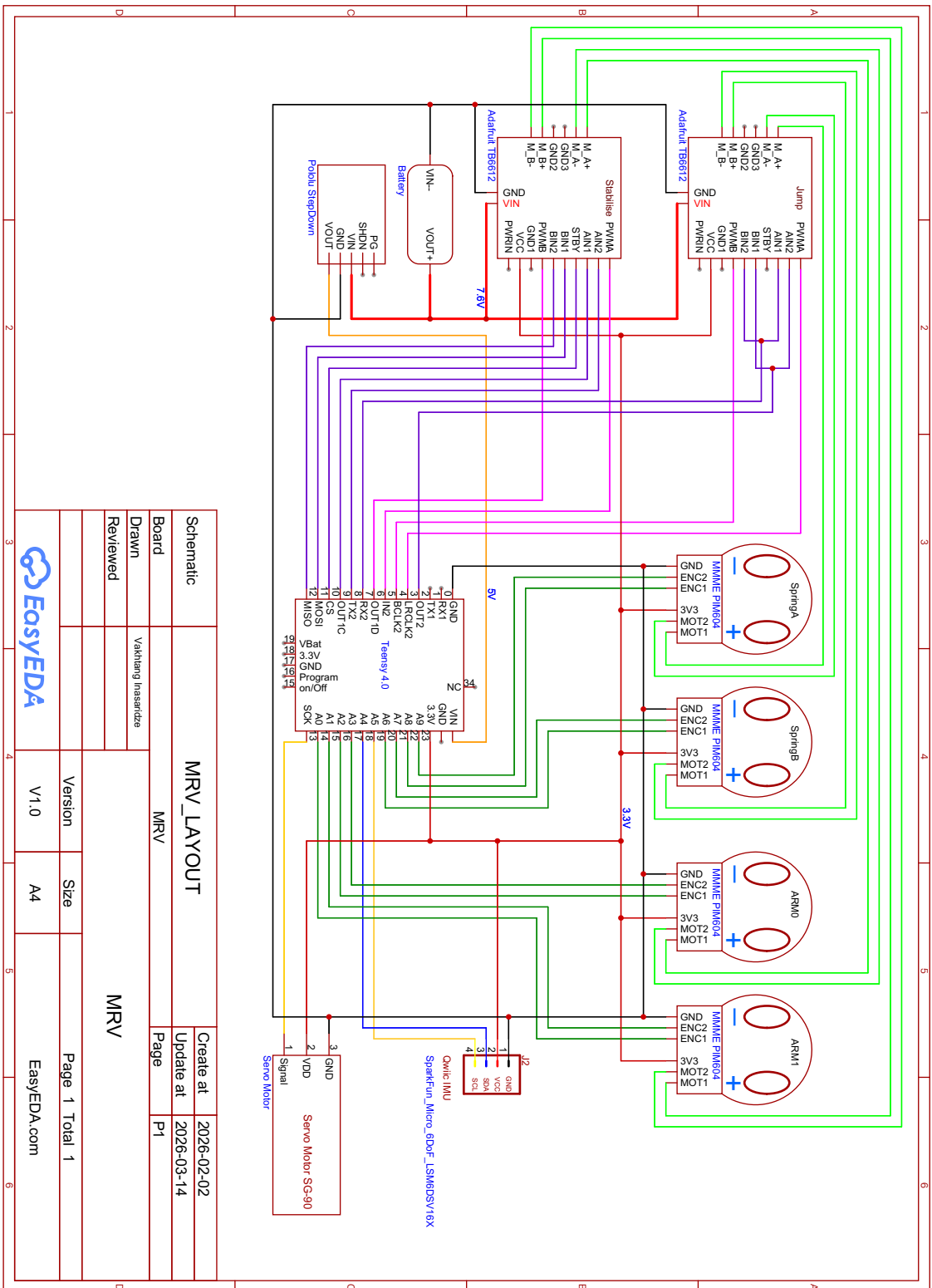
Relevant Appendices



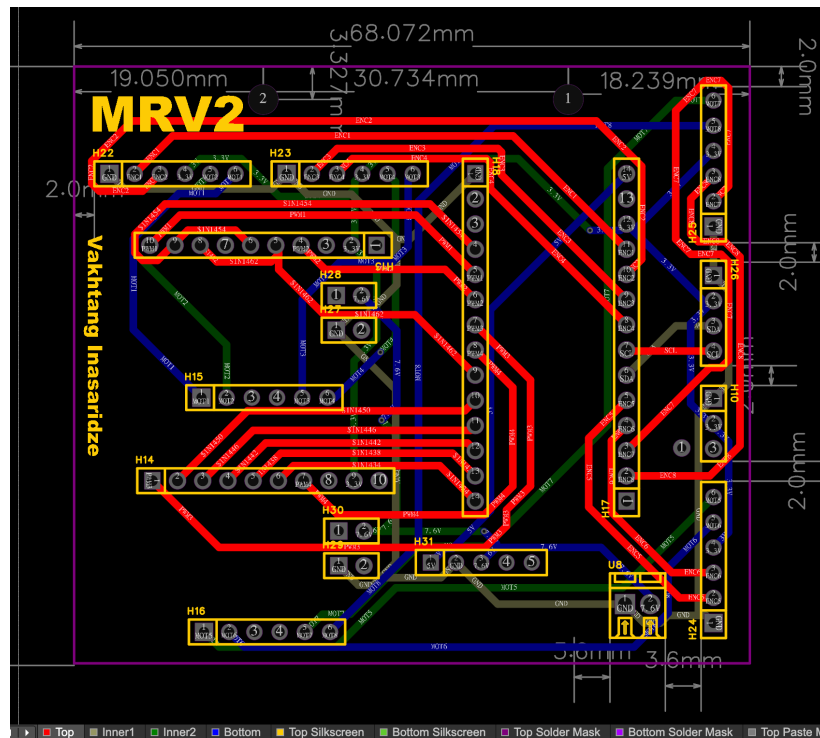
Appendix Fig. 1- Original Gantt chart from PFS, before initial exam break.



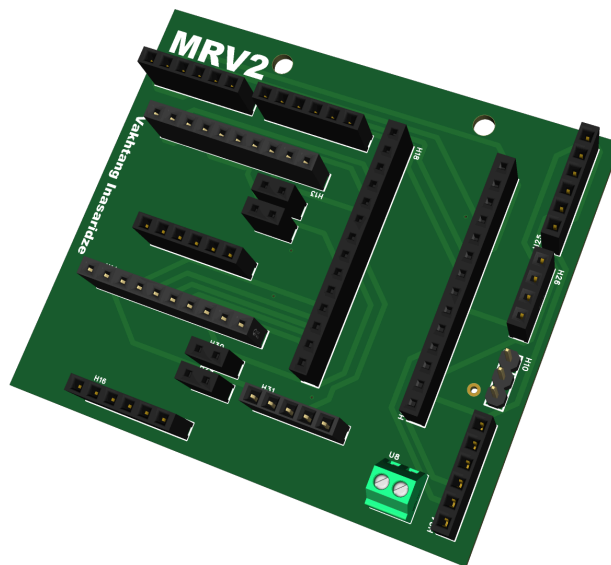
Appendix Fig. 2- Original Gantt chart from PFS, after initial exam break.



Appendix Fig. 3- Schematic of the MRV colour coded. Light green: motor wires, dark green: encoder signal, blue: directionality signal, pink: PWM, yellow: signal, red: power, black ground



Appendix Fig. 4- PCB schematic of the MRV. Blue: bottom layer, high voltage lines, green: ground layer traces, olive: motor output and misc., red: signal wires



Appendix Fig. 5-.3D render of the manufactured PCB from within EasyEDA. The PCB design used headers for modularity and DuPont connectivity.